

ActInf GuestStream 044.1 ~ Tsuchiya & Saigo: Category Theory, Consciousness

GuestStream | Tsuchiya & Saigo: Category Theory, Consciousness, Integrated Information | 1:39:40

Hello, I'm Hayato Saigo.

Okay.

Just one second.

I'll just make sure it's going.

It's all good.

All right.

Welcome to both.

It is June 13th at zero UTC.

We're here in active guest stream number 44.1 with Hayato Saigo and Nausugu Tsuchiya.

So really looking forward to this presentation and discussion.

Hayato, please say hello and then we'll continue with Nausugu's presentation.

Hello, I'm Hayato Saigo.

I'm a mathematician and a mathematical physicist also interested in consciousness.

I have worked a lot with Nao by using category theory and application to IIT or other consciousness theories.

So Nao.

All right.

So my name is Nao Tsuchiya.

I'm a professor in Monash University and also having a lab in ATR in Japan.

And as a sort of introduction to this today's session, I'll give a brief talk on our recent project called the Quaria Structure, which Hayato and I will be collaborating and also a little bit on the quantum Quaria hypothesis.

Both of them are hopefully relevant to active inference issue.

So the Quaria structure project is the Japanese funding that we received recently and it's going to continue for five years.

This will be to establish a super interdisciplinary research program to tackle the problem of consciousness, especially Quaria through the interdisciplinary research.

We are planning to also host summer school for two to three weeks every year.

And here's the fantastic members for this grant.

And Hayato is basically working with me in the part of the mathematical approach to the consciousness and the Quaria.

And so one of the problems, for example, we want to address is, for example, when you see this square, Daniel, what do you see here?

Blue.

Blue.

Yeah.

So you say blue and I say maybe owl in Japanese and the Hayato may say sunny in Russian, but we could mean the same experience, Quaria.

But how do we know that we are actually experiencing the same thing?

And I'm actually curious whether this kind of question is something that the free energy or active inference kind of style approach can ever tackle.

And my approach is to go into that category theory kind of inspired direction.

And here's the first study that published, you know, we published in a few years ago with Hayato and Shigeru together with also another category theorist, Steve Phillips, that in the category theory, there's an interesting lemma called Yoneza lemma that proves that all the relationship

characterizes its internal intrinsic properties and we are using that idea to study the Quaria.

That's the kind of inspiration of our approach.

So instead of, you know, trying to characterize what the blue is or red is, we are trying to characterize all the relationship between a particular color or particular Quaria to other exhaustively.

And then by finding the correspondence of this relationship, we try to identify what the actual blue is.

So if you will say blue is similar to green, but also similar to red, then that's going to be very different from my blue.

And that's the way how we detect this, you know, blueness is same or different.

And one recent application of this kind of idea is this unsupervised alignment that we are using with another collaborator, Masa Fumi, Oizumi University of Tokyo.

And I'll skip the detail.

And I'll skip the detail.

But basically what we are doing is to do some kind of experiment and then collect lots of data to obtain so-called similarity matrix between all the colors.

And then without really knowing what color maps onto what we try to use on the relationship to predict what we are seeing.

So in a sense, this is a purely category theoretical kind of, you know, approach to the problem of consciousness.

So then I'll just skip the details here.

And then, yeah, so we are doing this kind of approach on other types of the codec.

And then this is the thing that we are planning to do for the next five years.

And then another just brief kind of, you know, introduction.

If you have some other question, we can come back later.

So this is potentially more to do with that active inference aspect of our approach in a sense.

And this is called the quantum choreo hypothesis.

We published this preprint really recently.

And so you can access it later.

So one of the, there are many problems in the current, you know, models of our choreo or consciousness, where we tend to think that experience or consciousness is like, you know, one point in the high dimensional space.

And we use that kind of idea for our previous, you know, alignment experiment approach, but it lacks dynamics.

And it also lacks the, you know, effects that we tend to experience where, you know, the observation or report or introspection itself sometimes change our quality.

And that's sort of the active inference aspect that I'm talking about, you know, attention or introspection changes the experience.

But the traditional approach is very difficult to deal with this.

And one way to deal with that is this quantum form that we think is appropriate.

So here we think that, you know, there is some kind of a choreo experiencing system that is defined as this blue boundary.

You can consider that as sort of the Markov blanket like thing in the free energy kind of idea.

And then we consider a particular type of the quail, like redness or blueness itself as an observable.

It's a terminology in quantum theory, something that in principle can be observed within this, you know, aquaria.

And then we consider this measurement as a process that probes this, you know, aquaria's, you know, observable.

And unlike classical theoretical thinking, we think that the measurement or probing of this observable through its code of state changes its, you know, states and also observable itself.

And that's something that we can actually formalize mathematically using a concept called a very general concept of this conditional probability called instruments.

It's a, it's a very technical for the, you know, quick talk.

So I'll leave it about, you know, this is a modern quantum theoretical idea that captures both, you know, how we can affect on the world and that that quality itself can be affected by that measurement.

And then using this, we think that, you know, we can explain probably Claudia in a much more reasonable way with the mathematics, mathematical formalisms.

And so far, it's just a kind of theory, but we already think that it's probably better than the existing kind of a model of the, for example, attention and the consciousness.

And also, it also provides some kind of, you know, prediction and explanation of why some of the two concepts or two experiences can feel different depending on the order that you experience.

So this is just like conceptual kind of example.

But if I ask you, Daniel, that is China similar to North Korea, then you might say probably no, right?

I don't even know the reference point.

Yeah, okay.

But let's say, you know, just subjectively, you know, China doesn't feel similar to North Korea.

Okay.

Yeah.

Yeah.

However, if I reverse the order, is North Korea similar to China?

Then many people actually start to say that, ah, yeah, it's kind of similar.

Hmm.

Does that have to do with one being larger?

Or what is it about the order that primes people to respond differently?

So many people have already, you know, come up, came up with many different kinds of ideas to do, explain this kind of individual aspects of this, you know, order effects.

But as we mentioned, you know, we are actually computing or, you know, measuring this kind of the order effects systematically through all combinations of the colors, for example.

So, then it's hard to systematically explain why this order effect emerges in one case, but not the other and in which direction, right?

But that's what we found basically.

And that, using this, you know, quantum cognition kind of framework, we can explain it really nicely.

Hmm.

In this case, it's just a projection postulate, but that we are probably going to extend much more in the more instrument like, you know, our framework.

So, yeah, so, and also we can, we are also predicting that there may be some kind of violation, like a violation of inequality like situation in, you know, perception.

And if we can confirm that, that also, you know, gives a credence to our proposal, that the quality is like, you know, needs to be framed or explained in the quantum framework.

So I think that's probably enough to start up the conversation, I guess.

Awesome.

Okay, great.

Lot to jump into.

I'll share my screen so we can take some notes.

Okay.

Um, Hato, do you want to give a first remark?

Or I'm happy to, but, but do you have a first remark or what parts of the work do you want to reflect on first?

Hmm.

I think maybe the categorical approach, the importance of categorical approach is one issue to talk with you, I think, because you first pointed out that paper with Shigeru Tabuchi, 2016.

Um, so I'm, uh, interested in why you and your colleague, uh, are interested in that paper from the active inference site.

So, uh, can you explain some, uh, feeling or.

Yes.

Um, yes.

So this is, um, using category theory to assess the relationship between consciousness and integrated information theory with the two of you and Taguchi from 2016.

Mm-hmm.

So, um, I've long been interested in scale free or scale independent ways to describe systems and information theory has been recognized as one of those approaches.

Mm-hmm.

And information theory has been proposed to not just play, of course, a central role in the functionality of information systems.

Mm-hmm.

But with IIT and related theorems to also strike at the heart of the most quintessential aspect of our conscious experience.

So that seems like an incredible synthesis of the functionality of systems that don't necessarily seem to have anything qualia like at all, merely performing calculations like a super fast abacus basically.

Um, and so I am curious and feel free to address either of these things.

Mm-hmm.

When we describe information systems, what is it that would or could or is, uh, experienced?

Mm-hmm.

Are they experiencing their, their information, what they are processing?

Or what is it about information integration would grant a system the ability to move beyond a behavioral functionality into, again, strike at the heart of one of the most ineffable questions that the mind can really consider?

Mm-hmm.

Okay.

Interesting.

Um, actually the category itself, uh, in mathematics, originated, created in mid 20th century, uh, to bridge the two different fields.

That is, uh, geometry and algebra.

Uh, more specifically the topology, you know, something about topology maybe, um, which is a very, uh, general kind of, uh, mathematical methods.

So, uh, to capture the essence of the figures, uh, like, uh, uh, circle or the complicated, uh, figures.

And, uh, the fundamental problem of the topology is, uh, you, can you, uh, um, identify the two different, uh, figures?

Essentially.

That means you can, you, uh, can you, uh, transform continuously to the other figures or not?

So, if the one figure goes to another figure with a continuous transformation and back, then it is, uh, viewed as essentially the same.

But, uh, to, uh, to determine the two, uh, figures as essentially the same is a very difficult question.

So, um, like, great mathematician, uh, figured out the utility of the algebraic methods.

That, like, making the one figure to the other, uh, uh, from the figure, you can construct the algebraic system, like groups or something.

So, um, and then the, uh, the transform, uh, the problem into the algebraic region.

The, then the point is, very essential point is, uh, taking one figure to, uh, the, uh, constructing one algebra from the one figure is not sufficient.

sufficient. The point is the relation should be transformed the other regions. So one region here, other regions here, but one to one correspondence of objects is not sufficient.

But the relations, if you can transform the older relations into the other regions, then you can provide, you can prove something from one region to another region. Because this is a notion of the function, like the region, intuitively I talked about using the term region, it is mathematically called category. Category consists of objects and arrows, arrows or morphism, which is a general kind of relationship between these objects. So topological region is some kind of category of topology, like a top that objects are topological space and the continuous functions, continuous mapping is arrow or something. The other regions, for example, algebraic region, like groups and group homomorphism or the module vector space or something. So something algebraic and the relationship between them. So category is defined to formulate the intuitive concept of bridging something, functor. So first, the Eilenburg and McBrain define category to define this notion of functor by the bridging to region. Then, if this is established, then you can infer about topological region, topological category by using the algebraic relation or something. So we, that is, Nao and Shigeru and me talked about the possible application of this kind of argument into the consciousness study that it is just a problem of bridging different kind of regions, like physically testable region and our role, experience itself or something. So it is a apolio, it seems apolio. It is a too difficult question to have a problem to connect to these regions. But if you only focus on the object, but the relationship, if you can compare the relationship, then you can infer something about the, you cannot

directly touch you directly manipulate. So this is the idea.

Maybe I'll just add a little bit. So yeah, as Hayato said, you know, you are taking really diligent notes. So that may be quite good. But from my end, so this paper was inspired by really talking to Hayato and Shigeru and the meeting in this summer school on consciousness.

And there I encountered in this category theory for the first time. And there were several shocking kind of, you know, concepts that I learned from this first encounter. One is that, you know, there is a way to compare two completely different things, and then extract the common structure between them. And Hayato first introduced me this, you know, very strong, strange paper.

Which I didn't understand.

Which I didn't understand at all. But was so interesting that I could read, you know, for hours and hours. It's called the Rosetta Stone. It points out the commonality between the topology, quantum physics, logic, and computation.

Yeah.

Yeah.

Yeah, that was the title. And then I didn't understand what the heck is this in a paper, but that uses this category theory. And as a whole, I feel like, ah, I see. So maybe if we understand the structure behind this, each of this concrete example, we may be able to find a commonality or explanation at that level.

And that naturally applied to my, you know, program at the time, which is, you know, how to bridge the gap between the quadrilla and the information structure derived from neural, you know, activity. This is, this looks also completely different. But if there is any possibility to bridge the gap, it must be some kind of structure or mathematics. And that's this category theory.

Hmm.

know preserved.

Then it's much more than the usual kind of idea of a function between the sets right because it also imposed that you know correspondence between the relationship.

So in our idea basically if this is let's say Claudia of the category and if this is a neuronal activity or maybe information domain and then maybe you know we can have a back and forth of the factor that would be great.

Usually the most of the mathematical theory of consciousness tries to go from here to here right let's say in the case of IIT.

Arrow here and then what Hayat was kind of talking in a way as a natural transformation is this kind of you know secondary level arrow let's say I don't know whether free energy or active influence can count as a prediction from you know neural activity to the particular quality but let's say if there is something like that and then maybe there is some kind of a translation between these two theories.

Yeah if that exists that's called a natural transformation.

Awesome well a few comments on this the idea of mapping the maps between the spaces is so important it reminds me of metaphor where we're conveying relationships and saying well it's it's like a hammer or it is a hammer for this other situation so that's how we get few shot learning and applicability and how we grasp concepts is perhaps developmentally or from an evolutionary perspective in a large extent by generalizing our spatial and embodied understandings and then certainly communication if you want to have some semantics that are being communicated across minds through an interface through a classical interface then certainly you want the structure to be preserved and then I think the one that you have here is one of the most provocative for those empirically studying uh consciousness because it's kind of like this side is empirical and let's just say we don't know what the q side is for now but let's just say it's another space that isn't that and then if there is a certain uh landscape that neural activity patterns flow through like some kind of winterless competition or you get some sort of limit cycle and you find that there's a limit cycle of neural activity and then there's also uh that is being paralleled by experiential cycles then that is not just an accounting system that's a very strong finding about potentially the real underlying causal system especially because it can suggest unique other explanations or manipulations to make right do you want to mention anything about the metaphor

uh yeah yeah and uh before that uh very hard to say the technical remark is that for the original uh problem that connects um topology and algebra then uh in that case the algebra is more concrete more manipulatable in that case and the topology is something mysterious in this case so in general the factor is a kind of um generalization of mapping in sets so there's something mysterious into something concrete so now's uh picture c and d should be switched maybe in that case and yeah but uh of course there is a back and maybe back and forth so the idea is this our your blue and my blue program is for example you can prove that you can uh your blue and my blue is different if there's something uh informational or something the detectable structure is completely different then

ah your blue and my blue is different of course you cannot uh uh uh prove this the sameness between you you you are blue and uh my blue uh uh if there is no how to say the back and forth function but even if you don't have the uh the backwards uh function at least you can uh demonstrate that your blue and my blue is different seems to be different if the correspondence is functorial or something so uh it is a big step from the experience and the outside the under uh always me and the colleagues uh do such a thing so you are blue my my you are but this is uh not uh by focusing your blue my blue my blue only but your blue red something or you don't have to say even do not have to say the name of something so just uh how to say no labeling for example but the the relations you you can teach the relationship that these things you can teach the relationship these things so then compare that so very interestingly uh in general the structure is surprisingly uh uh coherent uh between persons persons or yeah so it is very nice and also maybe it's it is suggesting that the non-isomorphic case uh then uh your blue and my blue is something different or something so i think this approach is very interesting even if um the no bi-directional only one direction then but even that case it is useful yeah it is my remark

now if you want to yeah please please and then i'll have a question about your drawing right so this is uh one of the slides that i didn't show today but uh uh what what had was talking about is um you know kind of you know explaining this figure like uh the way you experience red is related to with the other experience uh you know not only color but shape and also in sound and so on in a very very complicated way and then we can actually measure how similar they are by asking the people and

then you can also do that for the other people and then these these two similarity matrix can be directly compared with our you know new method of this unsupervised learning and also iit for example may allow us in in the future to go back and forth between these you know our quaria structure and the information structure and then mathematically we believe that the information structure to information structure should be potentially you know translatable in some way and if that is possible in some way you know eventually we may be able to say ah okay so the way you are experiencing red is same or different in this way or that way that that's the sort of idea can you go back to your drawing right

awesome

so for the top arrow as we're if we're kind of just exploring what what might it mean that iit is going back and forth and also that active inference is going back and forth so what exactly does the edge between c and d represent in iit is that a quantitative evaluation like the five variable or what is being accounted for with that arrow

i have to do you want to say something yeah uh there is a various possibilities actually for example the sameness or difference or so if you have some kind of uh gradation of sameness so yeah sameness gradation but to a former uh to to formulate the gradation gradation of uh sameness you have to move on the enriched category not a simple kind of category but it's a generalized kind of category called enriched category in that case for example the simplest example of enriched category is a generalized uh matrix space but the like uh uh uh in matrix space for example the uh the point p and p itself has a

distance zero

of course of course of course and uh kind of uh p to q q to r uh is uh uh there is a triangular inequality or something so yeah

so yeah then it can be considered as a generalized kind of uh compositional structure called enriched category it is a typical kind of enriched category

uh so but the symmetry does not uh always hold or something so this is a one how to say the uh

uh a small step to grasp the our core structure because uh like uh uh now pointed out north korea china

distance and the china north korea distance is uh in general different or something so almost all

uh studies uh assume that some symmetry but uh we need to go go uh beyond that and uh then

maybe the one number or something is too poor to represent everything so more you have to go more and more

richer structure like uh not just numbers but some kind of projection operator or something then

you enter into the quantum region so relationships we feel should be maybe should be uh represented

from the some kind of the uh relationship between the notion of state uh in quantum quantum theory

and the state transition between that so our approach is step by step and you can choose

uh the many arrows uh to study something another something and then goes to unify the more and more general kind of arrows and categories and i i this is just for the speculation but

uh this kind of uh this kind of quantum theory theoretical like as a mathematical framework and not do

do not need to the kind of physics but the kind of mathematics uh can you use uh unify these kind of things

awesome just a few comments on on that when we're doing free energy minimization we use the kl

divergence which is very importantly not a distance because a distance would have the symmetry characteristic

and so the kl is chosen because it's easy to fit uh optimization

on these variational distributions but it's almost like it's a feature not a bug

that the kl divergence between q and p is not the same as p and q

because it turns out that that that kind of a of a divergence based um uh

quantification actually has some advantages um over a distance measure and then the mapping between

observables and hidden states and the cognitive operators that we bring to bear on that kind of

mapping problem those are some of the most fundamental pieces of the partially observable

markov decision processes that are used in the active inference generative model which is first the

mapping between observables and hidden states and then how do hidden states change through time and how

does action selection policy selection influence how hidden states unfold through time so this mapping between

hidden states and observables and between hidden states and the hidden state at the next time point

are and how again how action influences that if you understand those operations then it's like a game of

chess and the ball just rolls downhill and the variational distribution works exceedingly well but when those are off

base then you're fitting your way like right off a cliff

so uh maybe then i'll just add one uh more point so you you asked uh in the original like uh five minutes ago uh what this you know each of these points or arrow correspond to in the case of the iat uh and so on and then this is kind of related to all these uh ideas but uh one of the uh you know fundamental uh insights uh from iat but also from philosophy especially uh some kind of uh philosophy called the process theory is uh it is uh important to really understand that the process of flow or movement is really fundamentally uh important kind of uh important kind of aspect of things and then um that may be really distinguishing some kind of static sets or um the thing uh in the static world now when they start to interact and then when the process emerge then something uh starts to potentially you know uh generate uh or support uh subjective quality like you know uh proto-quartal kind of thing and uh in the case of the iat uh it is uh deeply kind of you know uh entrenched in this you know formation of the transition probability matrix of an entity like how this thing neuron or logic gate

it might have come from the previous state and it may go into the next state so any kind of a moment is already embedding the possible previous and the possible future there is some kind of a with us in a moment because it's a process and when the process is considered then you know uh as i have said that you know this kind of you know dot is a very you know kind of you know two thin kind of structure to think about so we need a more elaborated structure of a dot itself but the category theory is great uh in a sense that that each dot can itself contain complicated structure it's just you know uh uh represented as a single dot but it can contain something more right so the in an upshot one dot could correspond to a collection of the neuronal you know states and the connectivity pattern and things like that which can go from one to the other and then that may that this kind of movement may correspond to something like you know we see redness or whatever in this way but that's also with respect to other kind of you know arrows as well and that that's the sort of the wave of the relationship that we are trying to characterize both in the domain of aquaria but also brain is also the wave of the relationship gigantic ones right and as you said that you know transition from one state to the another state that's pretty much like dynamical state uh that dynamical system like you know approach and then uh the simple idea called monoid is uh uh applicable to this kind of dynamical uh system approach uh correct me

hayato if i'm saying something you know out of the bounds of my comfort zone but uh you know um any kind of a state can transition into another state and then this transition itself can be represented by the arrow the arrow and uh yeah so uh and that kind of combination of that particular kind of arrow composition may correspond to one experience also so that that's uh uh one thing i wanted to say the other thing that hayat also uh towards the end mentioned about the observable and states and why we are interested in the quantum uh uh you know framework quantum theory by the way it's not a quantum brain hypothesis we are not

saying that you know quantum you know phenomena in the brain is critical for consciousness but it's just a mathematical structure of the quantum theory that is important to understand the quality and the essence is most likely to be this you know non-commutativity

any kind of process that you do like you know cooking you know putting the water and then uh

model is different from model and then putting water right and so any kind of process we do there is some kind of temporal dependency that commutativity doesn't hold and that's also probably the case for the quodil and if that's the case uh then for you maybe it's a bit surprising but you know bayesian kind of statistics and bayesian probability may not work as you might wish yes so the in other words using the kind of mathematics there is a quite good generalization of bayesian uh innovation uh innovation uh kind of uh influence yes uh actually the the the one step generalization is called a projection uh in quantum mathematics that are then there is more generalization called uh instrument so instrument is a general notion of measurements like a system and the environment interacting the then uh more precisely the measurement is something that like the interaction which has the uh the uh the uh the uh apparatus is some very small tips or something which can be considered as a quantum the discount system and this probe system interacting quantumly then we can measure this probe by the macro side of the apparatus so this is a very general kind of uh measurement uh which is called uh which is identified mathematically with the notion of uh instrument originated by the davis and lewis and the uh the uh the uh one of the uh one of our collaborators uh beginning we begin in the collaboration but was our uh who is also uh famous for the ozawa's unique inequality was our inequality as a generalization of heisenberg universally valid version of uh heisenberg's inequality but uh he uh identified that this uh general notion of measurement and the instrument called completely positive in instrument in mathematics but this cp completely positive instrument is quite general and uh generalizing the very beautifully the bayesian influence uh into the quantum side so i think the the using uh focusing this kind of uh instrument kind of concept is the key maybe the tree uh to bridge the um after the point side on our side and quantum side or something so yeah it is my speculation but i think it is very fruitful direction oh it's awesome just one comment on that the the probe with the instrument or the quantum reference frame i believe as chris fields might describe it is an experimenter's choice and so the way that we could talk about policy selection and inference on which experiment to run would be in terms of which sequence of observations is going to minimize expected free energy okay so which sequence of observations is going to have high pragmatic value and epistemic value that's the expected free energy functional now as researchers we might seek out the most informative experiments and not prefer to see a certain measurement so we can say well all measurements are a priori equally pragmatically valuable to us so now expected free energy is going to be driven by information maximization just by pure learning but this recognition that action is woven into the process of reducing uncertainty about other systems across blankets it's it's absolutely fundamental because if you don't have the constraint that you need to actively select which experiments to run and to probe and to interfere in this quantum way this way described by quantum mathematics you're kind of in a delusional alternative world where like you get information from photons but you don't have a sensor so it's not an empirically tractable world unless we take on this constraint and then once we do our mathematics mathematics are the quantum mathematics and again it's kind of a subtle point this isn't the orc orr this

isn't necessarily taking a stance one way or the other on what kinds of electronic scale quantum phenomena exist in any system but rather just sees quantum mathematics as fundamental to decision making and uncertainty

which is quite a change from how it's been discussed for a hundred years

actually um yeah as now said we're not um depend on the hypothesis like uh the there is a quantum physical process uh behind the core of the process behind the core but why we can uh use the quantum mathematics is very very

simple reason because uh conventional probability theory classical probability theory uh depend on the probability space classical one is too narrow to capture for example the order effect or something

but uh you can generalize it so quantum mathematics is just generalized the classical mathematics and i think i believe that the why quantum mathematics is found for the kind of physics is very simple reason because in classical physics the some kind of the

uh determinism the classical determinism is very universal and the the probability only comes from the some kind of ignorance ignorance of us but in quantum physical uh context there is some kind of very fundamental uh randomness the fundamental uh accidents or something is found and they are all surprised because

if you go into the micro side the determinism should be stronger and stronger they thought but actually they are needs some kind of they are they found a fundamental non-commutativity and the macro micro side but uh

uh whoever bohrer and heisenberg or some people pointed out that uh even in classical or even in uh sorry uh macroscopic side but uh this kind of uh the importance of non-commutativity is very universal not especially in micro side but it is a historical reason that we first find in quantum mathematics in the micro side but

there is many many phenomenas uh which cannot be captured by the determinism or the classical probability theory

so i think the core or something is just some kind of yeah

example yeah example yes uh by the way maybe daniel can you uh replace this quantum consciousness with the quantum quaria yes so the quantum consciousness has been used to you know uh imply this quantum brain

hypothesis which may be you know potentially you know uh true but uh so far we don't really see the strong evidence uh about you know this um particular thing so

um

what's funny i i'm thinking of the classic qualia question what is it like to be a bat

bat and it's like what if the like is interpreted as a metaphor

what if there's a metaphorical mapping to bat which is an observer dependent mapping amongst space space uh

you know amongst spaces because we've moved the focus from the the objective

uh standalone experience of the bat to considering mappings

amongst cognitive systems one critical one being for example whoever's listening to the question and a bat

who's the one who actually answers that question

uh

yeah

oh okay so you you asked also about uh you know the inspiration from the category theory uh relevant to the um issue of the metaphor right and also communication and in fact uh so now it's a bit you know far from this you know our quantum corey and then jump back to uh another topic but uh so that metaphor uh uh model of a metaphor is something that the higher has been working on so i'll read uh yes yeah but the the other part which is this in communication

and that's something that we also wrote together uh last year um and using this you know concept uh called you know adjunction in uh category theory which may be also interesting to uh discuss so in 2016 paper we already mentioned that you know uh category theory is uh very promising for the study of consciousness precisely because it provides different levels of uh sameness

consciousness when i heard this from hayato i was like what what what do you mean but you know simultaneously it makes a lot of sense and in fact many many many you know controversy or you know confusion among the you know field of consciousness but not only for consciousness other fields like ecology as well like you know uh classification of the animal species and things like that that all up depends on this you know what do you mean by the thing

right and this is uh something that also relates to what you are talking about a moment ago which is observer dependence right so in the uh at the even at the really very bottom of this you know uh reality i think that what matters is some notion of this subjective choice of what it means to be the same equivalence or adjunction is a particularly interesting kind of situation where for example you know uh integer and the real number is definitely different but in terms of the relationship if you start to think about the relationship there is some kind of you know uh one-to-one correspondence between them and also uh hayato's book also uh in written in japanese but uh recently i read it and one of the interesting example is also multiplication and the exponentiation they you know at the level of object or number it doesn't you know match right but there is a very nice relationship between them that is very fundamental and something like this you know apparently completely different thing but finding really uh meaningful relationship that's very fundamental in communication you know we are talking different kind of language with different kind of meaning but because when i say something you reply and then there is some kind of systematic difference between what i expect and what i hear that's the kind of prediction error right i think the prediction error may not need to be minimized in fact but as long as it's consistent and systematic then it actually works as a you know very nice translational kind of a device like i don't have i continue you know after living like 20 years more more than 20 years of living outside of japan i still have very strong english accent either in japanese or you know japanese living in australia or you know uk us i can't minimize this prediction error but because i have this systematic prediction error for you you totally understand me right something like this is uh the uh interesting thing about that junction

well this this makes me think about uh nested reference frames in communication and let's just say that the the expected free energy or maybe even the integrated information content is such and such a level uh but in it varies within some range it's 100 plus or minus 10 instead of just trying to naively minimize that to zero we can create a nested model such that we expect that to be 100 plus or minus 10 at which point the non-zero value is stabilized or buffered and so then that becomes the kind of stabilizes or wraps the fact that we're not perfectly transmitting information but we can be perfectly aligned that we're not perfectly transmitting information and also our language understanding for fluent speakers we're able to map the phonemes a little bit deeper we're able to read between the lines or listen between the phonemes so that people who say words differently map to the exact same syntax let alone the semantics the sound gets kind of abstracted out but it influences our qualia so something is being passed forward with like tonality and the timing and i wonder about with of course large language models how much context they're able to garner from the string of tokens when for our experience sending and receiving natural language communication it's all about the timing and the tone and the setting and so finding ways that we can describe that that type of information the the category theory capacity for the different levels of sameness seems like exactly the tool that we need

i have to do you want to add anything yeah i'm thinking about that about that you know um the multi-level kind of free energy minimization i i actually have a question to you actually so i recently uh discussed uh with uh roboticist in japan uh jun tani yeah do you know his work yes who uh yeah is uh using this predictive coding kind of scheme uh for um constructing the self you know autonomous robot and uh he thinks this you know uh intermingle of the top-down prediction and the bottom-up you know our prediction error is the place that you know coria emerges and things like that but one of the kids you know most interesting uh aspect of that his model to me is that he has this different time scale for his uh neural system the most peripheral motor level and sensory level has a very fast you know time scale so that tries to match with you know what to have what's happening outside and then uh uh what they want to do towards you know outside and the intermediate level has a slightly longer kind of you know time scale so there is some kind of inertia and even higher level has a much much longer time scale and uh so in real probably neural system there is also different level of the time scale and also different levels of the mechanism acting at the same time like uh change in the activity state in uh spiking or not and also release of the uh synapses uh uh synaptic transmitters that could increase or decrease over the time and also even the new neurons emerging and are going out so uh um the the minimization when you are talking about minimization probably that kind of minimization is actually happening independently and also coherent entry potentially or contradictory across different levels and also regions and there may be some kind of effort or whatever to also you know coordinate these kind of contradictory demands of the minimization and uh what do you think about this issue do you still think it's an uh useful way to think about that or use the term minimization in this situation

that's a great question well one thing that stood out to me was you described the slower moving levels as having inertia and from the bayesian mechanics perspective that's exactly how we would model it they're slower moving they have more inertia and they move more like classical objects whereas the faster time scales are moving more like lighter particles more like that quantum probe in the limit and then this sort of

is the center or the highest nested model would be like the most cognitively massive potentially the slowest potentially integrating information over the longest time window and certainly this is uh underlain by these multiple time scales of mechanism spiking neurons from a few hundred milliseconds time scale neural modulatory processes over seconds and and minutes synaptic plasticity even uh neurogenesis and and our the

development of our brain so what does it really mean that minimization is occurring when we have this heterarchy where we have similar types of free energy minimizing units or active inference units within a level like

cortical columns or um grid cell units and then also we have these nested models that are uh optimizing across

uh spatio-temporally separated scales well one paper that looked at this was um our live stream 42 on the slam

simultaneous localization and mapping also in robotics and so they had a two uh nested model with two steps and the lower level was like the tactics and that was where you had the motor and the sensors and then the nested

model was slower that was like where do i want to be that was more like the strategy and um

you can create an expected free energy functional that sums across those two levels

how you tune the parameters is going to matter because uh how do you weigh yeah different levels yeah um

like and one what this comes up with like preference learning like if you have if your preference if you have two left or turn and right turn and your preference vector is like two and one versus two million and one million then that is the ratio of the preference is the same but the amplitude of the preferences is a lot

different which is going to swing the model towards valuing pragmatic outcomes whereas if you had 0.02 and

0.01 the ratio of the preference would still be the same but it's such a small amplitude preference and

so similarly with expected free energy minimization you could make it so that the the joint

minimization was driven by just the top level of the model and then the sort of contradictions

or even increases in expected free energy of the lower sensory model would basically always be accepted

in service of the top model or vice versa and it seems like biological systems with their sort of precarious

self-organized or self-organized criticality um operations they're poised where you can have like propagations where small changes in the bottom up model can propagate and um and induce like phase changes of higher levels

and vice versa so they're certainly poised parametrically at a special place where the composition of the faster sensory motor models and the slower like cognitive or narrative or metacognitive levels

it's all poised in a very integrated way potentially even one that is um where where the iit measures

uh tell us something useful um

um but yes the the contradictory demands
of subunits within a level of collective behavior and across levels
as a biologist i think we just point to ecology evolution and development and just distributed systems
that don't exist at an adaptive point or uh implement adaptive dynamics amidst uncertainty they just fail
to exist and so the complex biological systems that we observe today do manage that trade-off and then
in robotics they have this unique challenge of developing systems with one shot or with iterated
development that do have some of those properties but for sure there's no reason to think that just by
laying out a nested active inference model that you would actually get like a more adaptive decision
making agent in fact as you nest models the state space of these models becomes vast and with some back
of the envelope calculations you can see like well if i do this level of nesting and have this kind of
sensor array we need this absurd amount of data to actually parameterize this model and by the time we
collected that many observations maybe the causal structure that we learned to the world you know 200
hours of video ago don't apply
yeah i mean that that could be a quite a huge difference between uh probably iid and free energy
i i think um the iat uh explicitly states that uh this you know time scale that matters
but uh in this in this case it's uh just about you know quaria is a particular time scale that maximize
integrated information and that corresponds to the time scale of our experience like you know tani's
you know model you know that um slowest uh uh time scale is like on the order of 100 seconds or
something like that and also you know new uh synaptic plasticity is on the order of hours and uh most
likely spiking neurons seems to be more mattering with respect to our particular experience at each
moment right and why it is the case um that the provisional answer provided by iat is that they
because at that particular time scale integrated information is maximized and across the scale and
across the regions but this is a huge conjecture and nobody knows whether this is true or not
and uh when we use the fly model uh we weren't be uh we weren't able to sort of confirm any kind
of a clear peak in the integrated information um using a different time scale so it's still you know
something that i'm very interested in um looking but uh still not uh clear whether this is the case
i don't know if um hayato has some um idea or um insight into the uh appropriate time scale for
consciousness or something from the physics point of view yeah yeah yeah i i have some thoughts but
maybe that it becomes a long story so and also connects to the um observer depend dependence uh
problem so i i was think thinking about how to explain these very important things in one thread and i found
that
first um the very uh simple fact that there is no uh there comes no identically same two events
events uh every events is different in some respect but to do science or more generally some recognition of
law we need some the criterion of sameness for example you want to know how to feel at how to heal the
disease
uh you should uh identify the different persons uh people uh as the same same kind of uh the samples or
something of course there is no identical person but uh you should you have to uh identify the the same
in the members of the same in the members of the same group something so all the scientific studies is

dependent on some choice of the sameness and so in that sense the all the science depends on this kind of choice of the sameness uh as you both uh pointed out and uh there is no i think there is no canonical choice of the sameness you can choose every sameness but if you choose some sameness then uh you can talk

[illegible]

theoretical idea to the experimental thing is the key is expectation value.

So theory provides the expectation value of something.

And we can connect the theory and the experiments by comparing the theoretical expectation value and the average of samples.

So all the quantitative science depends on this structure like algebra of observables and some mapping from the observables algebra to the sum number system.

So this is called the state in mathematics.

And algebra and state is something like the mapping of quantities or random variables or observables into its expectation value.

So this mapping of expectation is technically called state.

So if you start from observable algebra and state, this is a starting point of non-commutative probability theory.

That is, if this algebra is commutative, then the conventional probability theory can reconstruct from this commutative algebra and state.

But if this is non-commutative as now pointed out, if the process is essential, then this algebra should be non-commutative.

Then we have the non-commutative probability space, which is, which I called intuitively the quantum mathematics or something.

So in this case, the observable dependence is described, the other kind of observable dependence can be described because you don't, you can't have the classical probability space for the, all the observable.

But if you choose one observable, one observable, like position or momentum, you can choose this, then you can construct the classical probability space.

So the choice of context or the choice of measuring something is by choosing something.

So context, you can have the classical kind of reality.

So this observable, this kind of observable dependence is well formulated by using non-commutative probability space.

So I think that these multi-layered observable, yeah, observable dependence is, can be described by using non-commutative probability.

The last thing I would say is the connection of the category theory and non-commutative probability. Very interesting.

If the category is given like a causal system, causal system give us some category like the causal variation is arrow and the composition is A is cause of B, B cause C. Then indirectly, of course, at least, the A cause C or something.

So it becomes a, can be considered as a category.

There is many, many kinds of categories.

But if the category is given, then the algebra reflecting this categorical structure is necessarily non-commutative probability.

Non-commutative probability is, from the mathematical point of view, it's kind of the two aspects of the same thing, like the fundamental observable dependence or contextual dependence or something. And this contextuality, like in topos theory, the actually this choosing concepts and the transformation between the context can be formulated in terms of topos, which is a very powerful kind of category. So, sorry for the too long explanation, but without that, this very fundamental question cannot be organized.

Sorry for the long, long explanation.

Oh, it's, it's awesome.

I mean, empirically, if you have a sensor that's getting measurements at a certain timescale, then there's a binding of the timescale of the, of the observer and the processing.

Can I ask a question from the chat?

All right.

Dave asks, at one time, Professor Tonini was representing the shape of the relations between complexity and consciousness, such that two richly connected systems are non-conscious.

They're overwhelmed by the complexity.

Recently, the inverted U that shows this relation seems to have become scarce, leading to undue confusion.

Have IIT people abandoned the idea that there is a bounded sweet zone for consciousness?

Can a system be either too simple or too complex to act consciously?

So, is IIT lead us to a kind of monotonic consciousness concept or could there be a U-shaped relationship with information integration and qualia?

I see.

Yeah, I can also see the comment here.

IIT lead us to a different perspective.

So, in the end, the question is, can the system be either too simple or too complex to act consciously, right?
In the sense.

Yeah.

From an IIT point of view, there are several things that need to be said.

But an inverted U is still there, in a sense.

And here, this person seems to be confusing between the axis of the kind of complexity and the axis of the connectedness.

So, in IIT, the original formulation, and it's still true for the IIT 4.0, which is the latest one.

So, when the system is, when the system is too connected, let's say, if you have, you know, hundreds of neurons and each of them are connected to everybody, and in a uniform way, then IIT predicts that it doesn't have much complexity or compositionality or uniqueness to arise from that kind of system.

And, you know, even if you cut the system, it doesn't matter much.

And so, that would lead to most likely very low level of consciousness.

I think that's what, yes, I should have typed connectivity.

Yes.

Yeah.

If that's the case, that's fine.

But higher complexity, which is granted by some small world or some sparse connectivity structure, higher complexity is monotonically associated with more consciousness?

That's not the case in IIT.

That's not the case in IIT.

And that is true from the beginning.

And the easiest kind of, you know, example is to have two highly complex kind of system that is loosely connected between them.

And it's a kind of a situation where you are interested, right, in a sense, like an ant colony.

Yeah.

Like each ant has a huge brain, you know, even though it looks extremely small, the number of the neurons they have is more than one million, right?

I think.

In that order, yes, around a million.

Yeah.

So IIT would naturally predict that, you know, the causal connection or integrated information arising from this one million neurons within the ant are going to be really massive.

It's very difficult to disconnect in any way, in any meaningful way.

However, interaction between a single ant to the another ant has a much less direct causal relationship.

And therefore, the local minima or local maxima of the five values is going to be very huge within each ant.

But when you combine or when you treat all these lots of ants, it looks, it definitely does, you know, bring some kind of intelligent and complex kind of, you know, behavior.

But phenomenality is not there at that level.

Phenomenality, consciousness, lives within this, you know, maximum level of, you know, integrated information, like pretty much like, you know, what we talked about in time scale stuff.

You know, we have a huge, you know, complex and, you know, very intelligent kind of behavior and artifacts in the long time scale and also short time scale.

But what matters for our phenomenology is this particular several orders of 100 milliseconds that corresponds to the maximum of the integrated information.

So you still have emergent intelligence at the colony level, but that doesn't necessarily mean that you have qualia arising at that level for analogous reasons why our qualia are at the spiking time scale, despite that being only one of the nested time scales.

Exactly, exactly.

Exactly.

Exactly.

Very cool.

IAT predicts that the intelligence and the consciousness qualia is completely, you know, dissociable.

And that, you know, leads to, you know, the commentary about the large language model, probably.

Right.

Large language model is probably structurally built to be very simple, almost like probably, you know, cerebral in the human brain, which accounts roughly like 80% of our neurons in the brain.

You know, out of 10 to the 11 neurons in the brain, you know, 80% are in the cerebral, but even if you, you know, remove it or, you know, lose it or bone without it, it seems, you know, consciousness is not that different.

And that's, and the simple kind of prediction that, you know, Cerebellum is doing maybe something similar to the, you know, large language model is doing.

It's very modular, it's parallel, it's not integrated, does a lot of intelligent things, but nothing to do with the consciousness.

Very interesting.

Well, maybe in our closing sections, of course, anything you want to share or ask, but how does active inference benefit and grow from what you're sharing?

what you're sharing and how do you see things unfolding for people interested in these topics?

I have to do you want to say something first?

Yeah, I was thinking how it now.

Okay, then from my end, so I used to have difficulty in thinking about the active inference or a Monaco blanket and things like that.

But, you know, by broadening my perspective and so becoming more generous, I started to feel like more, maybe some aspects of the thinking and also formulation can be relevant, at least for the explanation of the biological systems in various domains.

And if it's also useful to explain intelligence or construct intelligence, that would be great.

And from our work of the quantum quadrior hypothesis, if a quadrior is something that is influenced by its measurement, then that aspect may be, you know, emphasizing this.

Usually, you know, quadrior consciousness is kind of considered a passive.

You know, construct like at a given moment, what I'm experiencing, static, passive.

But now I'm more thinking about because of this, you know, collaboration with the higher-than other people, that can be the case.

And it's more to do with the process.

And if that's a process, it may not be that kind of the level of the active inference that you're thinking, like, you know, not the level of the acting on the world,

but more to do with that, this state itself, influencing itself might be some kind of important insight to understand the quality of it.

Okay.

Okay.

I, my comment is the, okay.

So the, it is related to the concept of states again, because expectation value, I pointed out the importance of expectation value.

But for example, what is the expectation value for the, for example, the cause, strength of causality or something.

Then the problem of timescale is very important.

So the, you can cause grain, for example, the brain activity for one timescale, another timescale.

So I think if IIT is correct, there is a good timescale, which is important, the importance of the, choosing the, this structure.

Then I think the brain information or something causal structure should be the good category and you can construct a non-computative algebra and state is defined or something.

So the timescale is quite fundamental.

And on the other hand, the state on the category, that is, state on the category algebra can be represented by some kind of weight on arrows, the positive kind of weight on arrows.

So the changing the state is changing the weight of these arrows, the positive definite function.

I think this, the transition or the weighting of the, this arrow is very, from, at least mathematically, very interesting to compare the active inference.

So, as a mathematical point of view, so, good weight, weight to the better weight or something is very, how to say, interesting to compare the, I'm not, I don't have much knowledge about active inference, but it seems to me, it is quite useful to connect to this level, the state change of or something.

So, yeah, it is very interesting to me.

And other, other thing is, I forgot the third point.

But, yeah, so, yeah, yeah.

And, yes, as now already, okay, point out.

The, I, I talked about the measurements, instruments, but the point of the, this notion is a probe system.

So, measurement is always, often considered as a very direct kind of interaction like the system and the environment.

But, environment has some kind of probe into the system.

Then, the interaction is, occurred with this probe and this system.

I think it is very analogous to the Quoria, the KSOL.

So, of course, outer world is relating to the brain inside, of course, at least the process or the historically or developmentally.

But, I think the, this, through perception or something, the brain has its probe of the world, for example.

This interaction is very hard to say, interesting.

So, measurement, the analogy to the measurement theory, maybe that some, physically within the brain, but some kind of probe of the outer system, and this Quoria is interacting.

So, I think the, it unifies the picture, I think.

Maybe, what do you think now, for example?

Yeah, yeah, yeah, so that, that's actually very much related to what I was actually constructing this morning.

Yeah, so I was reading this Ozawa paper that describes the instrument.

And I, I now, I'm getting better understanding what it means, but what he says is that every observable A of a system S is identified with the observable A tensored with the instrument I of a system S plus S prime.

So, S here is the original system, and S prime is the one that with any system S prime external to S .

And what the Hayato was meaning is that this S prime is a probe system, which is a part of the environment, right?

Yeah.

And so here, what this, you know, quantum, in this quantum kind of, you know, system description, but it's also true for the quantum cognition.

And in my case, it is also true for the quantum quantum hypothesis is that some kind of quantum quantum quantum system is interacting with the world, but through the sort of membrane, or sort of that, like a bank of blanket like thing, already filtering.

So, this is a world interaction with the sensory input and attention.

And this part itself is kind of, you know, interacting with itself.

Yeah.

These arrows are the critical part.

And then if we translate that, you know, other words, way to, in our case, this corresponds to every coil A of a coil experiencing system S , which is in a case of IAT, it's called a complex.

Observer A times instrument over system S plus S prime.

And this S prime here, I think will include all the neuronal processes that support the underlying coil.

Yeah.

But it doesn't really necessarily directly include the external environment.

Oh, yeah.

You know, many people start to do that.

But you know, if you do that, then suddenly, you know, hallucination or dreaming and things like that becomes very difficult to explain.

Yeah.

Yeah.

Including too much.

Mm hmm.

Mm hmm.

Mm hmm.

Yeah, I think that's correspond to what you wanted to say, right?

Yes, exactly.

Exactly.

Yeah.

Yeah.

Well, this is really awesome.

I very much appreciate that we could have this conversation.

Is there anything else you want to add or ask?

Yeah.

Thank you.

I also enjoyed and it was also good for me to formulate my thoughts as well.

Yeah.

I'm very surprised.

Yeah.

Surprised mean the.

the.

How consistent our thoughts are.

So how to say that it's.

It was just a surprise for me.

And maybe.

the conversation often goes, how to say, diverse, but our conversation has a very surprisingly conversion.

So I think it's quite interesting in itself.

And yeah, I run, I learn much from this conversation.

Thank you for inviting us.

Thank you so much.

So hope to join on your channel and hear updates on your research as things develop.

So thank you again.

Thank you.

Thank you.

Bye bye.

Bye bye.

Bye bye.

Bye bye.

Bye bye.

Bye bye.